Lesson 9: Probability Inequalities

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10 Probability Inequalities

Concentration inequalities bound how far a random variable can deviate from its expectation. They are the workhorses of convergence proofs.

10.1 Markov's Inequality

Markov's Inequality

Theorem 10.1 (Markov's Inequality). *If $X \geq 0$ is a non-negative random variable and $a > 0$, then:*

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a}. \tag{1}$$

Proof. Since $X \geq 0$, we have $X \geq a \cdot \mathbf{1}_{\{X \geq a\}}$ pointwise (the indicator is 1 when $X \geq a$ and 0 otherwise; in both cases, X is at least $a \cdot \mathbf{1}$). Taking expectations:

$$\mathbb{E}[X] \geq \mathbb{E}[a \cdot \mathbf{1}_{\{X \geq a\}}] = a \cdot \mathbb{P}(X \geq a). \tag{2}$$

Dividing by $a > 0$ gives the result. □

Remark 10.1. Markov's inequality is the weakest concentration inequality but requires the least assumptions (only non-negativity and finite mean). All stronger inequalities in this section are derived from it by applying Markov to a transformed variable.

10.2 Chebyshev's Inequality

Chebyshev's Inequality

Theorem 10.2 (Chebyshev's Inequality). *If X has finite variance $\sigma^2 = \mathrm{Var}(X)$, then for any $\varepsilon > 0$:*

$$\mathbb{P}(|X - \mathbb{E}[X]| \geq \varepsilon) \leq \frac{\mathrm{Var}(X)}{\varepsilon^2}. \tag{3}$$

Proof. Apply Markov's inequality (Theorem 10.1) to the non-negative random variable $(X - \mathbb{E}[X])^2$ with threshold ε^2:

$$\mathbb{P}(|X - \mathbb{E}[X]| \geq \varepsilon) = \mathbb{P}\big((X - \mathbb{E}[X])^2 \geq \varepsilon^2\big) \leq \frac{\mathbb{E}[(X - \mathbb{E}[X])^2]}{\varepsilon^2} = \frac{\mathrm{Var}(X)}{\varepsilon^2}. \tag{4}$$

□

Connection to RL

Chebyshev's inequality is used in the proof of the weak law of large numbers (Theorem **??**), which justifies Monte Carlo estimation: the sample mean of returns converges in probability to $V^\pi(s)$.

10.3 Cauchy–Schwarz Inequality

Theorem 10.3 (Cauchy–Schwarz Inequality). *For random variables X, Y with $\mathbb{E}[X^2], \mathbb{E}[Y^2] < \infty$:*

$$|\mathbb{E}[XY]| \leq \sqrt{\mathbb{E}[X^2]\,\mathbb{E}[Y^2]}. \tag{5}$$

Equivalently, $(\mathbb{E}[XY])^2 \leq \mathbb{E}[X^2]\,\mathbb{E}[Y^2]$. Equality holds if and only if $Y = cX$ a.s. for some constant c.

Proof. If $\mathbb{E}[Y^2] = 0$, then $Y = 0$ a.s., so $\mathbb{E}[XY] = 0$ and the inequality is trivial. Assume $\mathbb{E}[Y^2] > 0$.

For any $t \in \mathbb{R}$, $\mathbb{E}[(X + tY)^2] \geq 0$. Expanding:

$$\mathbb{E}[X^2] + 2t\,\mathbb{E}[XY] + t^2\,\mathbb{E}[Y^2] \geq 0 \quad \text{for all } t \in \mathbb{R}. \tag{6}$$

This is a non-negative quadratic in t, so its discriminant must be non-positive:

$$(2\,\mathbb{E}[XY])^2 - 4\,\mathbb{E}[X^2]\,\mathbb{E}[Y^2] \leq 0, \tag{7}$$

which gives $(\mathbb{E}[XY])^2 \leq \mathbb{E}[X^2]\,\mathbb{E}[Y^2]$. □

Corollary 10.4 (Covariance Bound)**.** $|\operatorname{Cov}(X, Y)| \leq \sqrt{\operatorname{Var}(X)\operatorname{Var}(Y)}$, *with equality iff* $Y = aX + b$ *a.s.*

Proof. Apply Theorem 10.3 to $X - \mathbb{E}[X]$ and $Y - \mathbb{E}[Y]$. □

10.4 Hoeffding's Inequality

Lemma 10.5 (Hoeffding's Lemma)**.** *If X is a random variable with $\mathbb{E}[X] = 0$ and $a \leq X \leq b$ a.s., then for all $\lambda > 0$:*

$$\mathbb{E}[e^{\lambda X}] \leq \exp\left(\frac{\lambda^2 (b-a)^2}{8}\right). \tag{8}$$

Proof. Since $a \leq X \leq b$, we can write $X = \alpha a + (1-\alpha)b$ where $\alpha = (b - X)/(b - a) \in [0, 1]$. By the convexity of $e^{\lambda x}$:

$$e^{\lambda X} \leq \frac{b - X}{b - a}\, e^{\lambda a} + \frac{X - a}{b - a}\, e^{\lambda b}. \tag{9}$$

Taking expectations (using $\mathbb{E}[X] = 0$):

$$\mathbb{E}[e^{\lambda X}] \leq \frac{b}{b - a}\, e^{\lambda a} + \frac{-a}{b - a}\, e^{\lambda b}. \tag{10}$$

Let $p = -a/(b - a)$ and $u = \lambda(b - a)$. Then the right-hand side equals $e^{g(u)}$ where $g(u) = -pu + \ln(1 - p + pe^u)$. One verifies $g(0) = 0$, $g'(0) = 0$, and $g''(u) \leq 1/4$ for all u (since $g''(u) = p(1-p)e^u/(1 - p + pe^u)^2 \leq 1/4$ by AM-GM). By Taylor's theorem with Lagrange remainder: $g(u) \leq u^2/8$. Thus $\mathbb{E}[e^{\lambda X}] \leq e^{\lambda^2 (b-a)^2/8}$. □

Hoeffding's Inequality

Theorem 10.6 (Hoeffding's Inequality)**.** *Let $X_1, \ldots, X_n$ be independent random variables with $a_i \leq X_i \leq b_i$ a.s. Let $\bar{X} = \frac{1}{n}\sum_{i=1}^n X_i$. Then for any $t > 0$:*

$$\mathbb{P}\big(|\bar{X} - \mathbb{E}[\bar{X}]| \geq t\big) \leq 2\exp\left(-\frac{2n^2 t^2}{\sum_{i=1}^n (b_i - a_i)^2}\right). \tag{11}$$

Proof. Let $S = \sum_{i=1}^n (X_i - \mathbb{E}[X_i])$. We bound $\mathbb{P}(S \geq nt)$; the other tail is symmetric.

Step 1: Chernoff bound. For any $\lambda > 0$, by Markov's inequality:

$$\mathbb{P}(S \geq nt) = \mathbb{P}(e^{\lambda S} \geq e^{\lambda nt}) \leq e^{-\lambda nt}\,\mathbb{E}[e^{\lambda S}]. \tag{12}$$

Step 2: Factor the MGF. Since the X_i are independent, the centered variables $X_i - \mathbb{E}[X_i]$ are independent, so:

$$\mathbb{E}[e^{\lambda S}] = \prod_{i=1}^n \mathbb{E}[e^{\lambda(X_i - \mathbb{E}[X_i])}]. \tag{13}$$

Step 3: Apply Hoeffding's Lemma. Each $X_i - \mathbb{E}[X_i]$ has mean 0 and lies in $[a_i - \mathbb{E}[X_i], b_i - \mathbb{E}[X_i]]$, an interval of length $b_i - a_i$. By Lemma 10.5:

$$\mathbb{E}[e^{\lambda(X_i - \mathbb{E}[X_i])}] \leq \exp\left(\frac{\lambda^2(b_i - a_i)^2}{8}\right). \tag{14}$$

Therefore $\mathbb{E}[e^{\lambda S}] \leq \exp\left(\frac{\lambda^2}{8}\sum_i (b_i - a_i)^2\right)$.

Step 4: Optimize over λ. Combining with (12):

$$\mathbb{P}(S \geq nt) \leq \exp\left(-\lambda nt + \frac{\lambda^2}{8}\sum_i (b_i - a_i)^2\right). \tag{15}$$

The exponent is minimized at $\lambda^* = 4nt/\sum_i (b_i - a_i)^2$, giving:

$$\mathbb{P}(S \geq nt) \leq \exp\left(-\frac{2n^2t^2}{\sum_i (b_i - a_i)^2}\right). \tag{16}$$

The same bound holds for $\mathbb{P}(S \leq -nt)$ by applying the argument to $-S$. Combining both tails with the union bound gives (11). □

10.5 Azuma–Hoeffding Inequality

Azuma–Hoeffding Inequality

Theorem 10.7 (Azuma–Hoeffding Inequality)**.** *Let $\{M_k\}_{k=0}^n$ be a martingale with respect to $\{\mathcal{F}_k\}$ such that $|M_k - M_{k-1}| \leq c_k$ a.s. for constants $c_k > 0$. Then for any $t > 0$:*

$$\mathbb{P}(|M_n - M_0| \geq t) \leq 2\exp\left(-\frac{t^2}{2\sum_{k=1}^n c_k^2}\right). \tag{17}$$

Proof. Let $D_k = M_k - M_{k-1}$. Then $\mathbb{E}[D_k \mid \mathcal{F}_{k-1}] = 0$ and $|D_k| \leq c_k$, so $-c_k \leq D_k \leq c_k$.

For any $\lambda > 0$, by the Chernoff bound (Markov's inequality applied to $e^{\lambda(M_n - M_0)}$):

$$\mathbb{P}(M_n - M_0 \geq t) \leq e^{-\lambda t}\,\mathbb{E}[e^{\lambda(M_n - M_0)}]. \tag{18}$$

We bound $\mathbb{E}[e^{\lambda \sum D_k}]$ by iterative conditioning. Note that:

$$\mathbb{E}[e^{\lambda \sum_{k=1}^n D_k}] = \mathbb{E}\left[e^{\lambda \sum_{k=1}^{n-1} D_k} \cdot \mathbb{E}[e^{\lambda D_n} \mid \mathcal{F}_{n-1}]\right]. \tag{19}$$

Since $\mathbb{E}[D_n \mid \mathcal{F}_{n-1}] = 0$ and $|D_n| \leq c_n$ conditionally, Hoeffding's Lemma applied conditionally gives:

$$\mathbb{E}[e^{\lambda D_n} \mid \mathcal{F}_{n-1}] \leq \exp(\lambda^2 c_n^2/2). \tag{20}$$

(The interval is $[-c_n, c_n]$, so $(b-a)^2/8 = (2c_n)^2/8 = c_n^2/2$.)

Iterating this n times:

$$\mathbb{E}[e^{\lambda(M_n - M_0)}] \leq \exp\left(\frac{\lambda^2}{2}\sum_{k=1}^n c_k^2\right). \tag{21}$$

Optimizing over λ as in Hoeffding's proof gives $\mathbb{P}(M_n - M_0 \geq t) \leq \exp(-t^2/(2\sum c_k^2))$. The two-sided bound follows by symmetry (apply to $-M_k$). □

Connection to RL

The Azuma–Hoeffding inequality bounds the fluctuations of martingales, which arise in TD learning as partial sums of TD errors. It provides finite-sample concentration bounds for value function estimates, complementing the asymptotic convergence guaranteed by stochastic approximation theory.

10.6 Jensen's Inequality

Definition 10.1 (Convex Function)**.** A function $\varphi : \mathbb{R} \to \mathbb{R}$ is *convex* if for all $x, y \in \mathbb{R}$ and $\lambda \in [0, 1]$:

$$\varphi(\lambda x + (1 - \lambda)y) \leq \lambda\, \varphi(x) + (1 - \lambda)\, \varphi(y). \tag{22}$$

Lemma 10.8 (Supporting Hyperplane / Tangent Line)**.** *If φ is convex, then for any $c \in \mathbb{R}$, there exists a constant $m \in \mathbb{R}$ (a subgradient) such that:*

$$\varphi(x) \geq \varphi(c) + m(x - c) \quad \textit{for all } x \in \mathbb{R}. \tag{23}$$

Proof. Define $m = \sup_{y<c} \frac{\varphi(c)-\varphi(y)}{c-y}$. By convexity, the slopes of secant lines through $(c, \varphi(c))$ are non-decreasing. For any $x > c$, the secant slope from c to x is at least m:

$$\frac{\varphi(x) - \varphi(c)}{x - c} \geq m, \tag{24}$$

which gives $\varphi(x) \geq \varphi(c) + m(x - c)$. For $x < c$, the secant slope from x to c is at most m:

$$\frac{\varphi(c) - \varphi(x)}{c - x} \leq m, \tag{25}$$

which rearranges to $\varphi(x) \geq \varphi(c) + m(x - c)$. The case $x = c$ is trivial. □

Jensen's Inequality

Theorem 10.9 (Jensen's Inequality)**.** *If φ is convex and $\mathbb{E}[|X|] < \infty$, then:*

$$\varphi(\mathbb{E}[X]) \leq \mathbb{E}[\varphi(X)]. \tag{26}$$

Conditional version: $\varphi(\mathbb{E}[X \mid Y]) \leq \mathbb{E}[\varphi(X) \mid Y]$.

Proof. Let $c = \mathbb{E}[X]$. By Lemma 10.8, there exists m such that $\varphi(X) \geq \varphi(c) + m(X - c)$ pointwise. Taking expectations:

$$\begin{aligned} \mathbb{E}[\varphi(X)] &\geq \mathbb{E}[\varphi(c) + m(X - c)] \\ &= \varphi(c) + m\big(\mathbb{E}[X] - c\big) && \text{(linearity)} \\ &= \varphi(c) + m \cdot 0 = \varphi(c) = \varphi(\mathbb{E}[X]). && (27) \end{aligned}$$

For the conditional version, fix $Y = y$ and apply the unconditional Jensen's inequality to the conditional distribution of X given $Y = y$. Since this holds for every y, it holds as a random variable inequality. □

Connection to TD Learning

Jensen's inequality is used in Appendix B of the article to prove the contraction property $\|\mathbf{P}_\pi \mathbf{v}\|_{d^\pi} \leq \|\mathbf{v}\|_{d^\pi}$. The key step: $(\mathbf{P}_\pi \mathbf{v})(s') = \sum_s \frac{d^\pi(s) P_\pi(s'|s)}{d^\pi(s')} v(s)$ is a weighted average

of $v(s)$ with weights summing to 1 (a conditional expectation). Since $\varphi(x) = x^2$ is convex:

$$\left(\sum_s w_s\, v(s)\right)^2 \leq \sum_s w_s\, v(s)^2.$$

This proves $\mathbf{P}_\pi$ is non-expansive in the d^π-weighted norm, which in turn shows the matrix $\mathbf{A}$ in the linear TD fixed-point equation is negative definite.